TCS and Jaguar TCS Racing Prepare for a New Chapter in Formula E with Digital Twins

Topline Messages: *Pushing boundaries in the virtual world to improve performance in the physical world.*

- **Challenge:** With Formula E races often taking place on temporary street circuits and tight regulations, Jaguar TCS Racing can't often practice on physical circuits or run in-person tests. Regardless, teams must still find ways develop top performing cars.
- **Question:** *Can a vehicle's on-track behavior be replicated in a simulated environment?* TCS and Jaguar TCS Racing set out to answer this question by enhancing the team's digital twin technology so that it can run more simulations, generate more data, and validate software code.
- **Solution:** TCS's automotive data science team developed a mathematical model, called the Virtual Vehicle Validation Model (V3M), which instructs the digital twin of the Jaguar I-TYPE 7 on how to navigate a track based on various inputs. The V3M allows Jaguar TCS Racing to: 1. Run continuous simulations without a person in the simulator. 2. Rapidly and robustly validate changes to developmental software in digital twin models.
- **Why This Matters:** Running the V3M saves Jaguar TCS Racing engineers critical time during race preparation. With the onset of the GEN3 Evo era, teams can make changes to the vehicle software right up to the very last moment in pursuit of performance gains – the V3M allows them to be perpetually adaptive at the factory and at the race track.

Jaguar TCS Racing Use of Digital Twin Technology

- **Advances in Formula E Technology:** Over the course of 10 seasons, the all-electric ABB FIA Formula E World Championship has experienced incredible advances in technology. The 2024/25 season is no different with the new GEN3 Evo era.
    - The Jaguar I-TYPE 7, driven by Jaguar TCS Racing, can accelerate from 0-60mph in 1.82-seconds (0-100kmh in 1.86-seconds), making the GEN3 Evo Formula E cars the fastest accelerating FIA single-seaters in the world.
    - With new bodywork, all-wheel drive capabilities for Qualifying Duels, the launch and Attack Mode phases of races, plus new Hankook iON tires with 5-10% more grip, Formula E racing will be faster and closer than ever before.

- **Sporting Restrictions:** With Formula E races often taking place on temporary street circuits and tight regulations, Jaguar TCS Racing does not always have the ability to practice on physical circuits or run in-person tests. Regardless, teams must still find ways develop top performing cars.

- **Simulation is the Solution:** Simulation and digital twin technology are fundamental for the development and optimization of the Jaguar I-TYPE 7, as well as for race preparation and strategy, allowing Jaguar TCS Racing to operate quickly, efficiently, and reliably.
    - These technologies allow the racing team to practice in the virtual world, moving beyond physical limitations and accelerate their learnings.

- **Driver-in-the-Loop:** The team's primary digital twin is a driver-in-the-loop simulator, situated at their technical headquarters in Kidlington, Oxfordshire. It's used across all areas of technology – from software development to driver preparation.
    - The simulator aims to virtually replicate key features of the Jaguar I-TYPE 7. Using the highest level of correlation to reality, this digital twin simulates the physics of the car on the racetrack when in motion, including the impact of torque, steering, and regenerative braking.

- **Digital Twin Difference:** Digital twins allow Jaguar TCS Racing to innovate and validate without the constraints of physical deployment. This is especially important for the software code used in the car at each race.
    - What is a digital twin? A digital twin is a digital representation of an intended or actual real-world physical product, system, or process and can be used for practical purposes, such as simulation, testing, training, and maintenance.

- **Better Simulation = Better Data = Better Performance:** Software code changes so quickly that it would almost require someone driving continuous laps in the simulator to find the bugs and discover the optimal code for each race.

- **The Big Question:** *Can a vehicle's behavior in a simulated environment be replicated?*
    - TCS and Jaguar TCS Racing set out to answer this question by enhancing the team's digital twin technology so that it can run more simulations, generate more data, and validate software code.

- **The Virtual Vehicle Validation Model:** TCS's automotive data science team developed a mathematical model, called the Virtual Vehicle Validation Model (V3M), which instructs the digital twin of the Jaguar I-TYPE 7 on how to navigate a track based on various inputs. The digital twin car consists of the car physics software and different modules.
    - The V3M allows Jaguar TCS Racing to:
        - Run continuous simulations without a person in the simulator.
        - Rapidly and robustly validate changes to developmental software in digital twin models.
    - When the drivers enter the simulator to begin their race prep, Jaguar TCS Racing can have complete confidence that the simulator results are accurate, unlocking creativity and innovation in search of performance gains.

- **Why this Matters:** With the onset of the GEN3 Evo era, teams can make changes to the vehicle software right up to the very last moment in pursuit of performance gains, therefore, building confidence in software is more important than ever for Jaguar TCS Racing.
    - Running the V3M saves the Jaguar TCS Racing engineers critical time during race preparation by rapidly validating changes.
    - *Once possible, Jaguar TCS Racing to provide stats on how much time the V3M saved / how it is improving the setup.*

- **What's Next:** In the future, the V3M could be integrated with artificial intelligence to provide even more insights into the performance of the car digital twin. The lifeblood of AI is strong and reliable data. With the support of the V3M, Jaguar TCS Racing can count on accurate simulation insights.

- **Race-to-Road Technology Transfer:** The advanced data tools and techniques TCS uses to support simulation for Jaguar TCS Racing have evolved from TCS's Connected Autonomous Vehicles (CAVs) test bed. In turn, TCS have the potential to utilize the learnings from an extreme racing environment to improve the development of road cars, creating a cycle of continuous race-to-road technology transfer.